

Ismael MARKRIA

Mathematics student at EPFL — Python / C++ development, data and scientific computing

Lausanne, Switzerland

github.com/imarkria · linkedin.com/in/ismael-markria



PROFILE

Mathematics student at EPFL, comfortable moving between theory and code. Object-oriented programming in Python and C++, solid foundations in linear algebra, optimization and numerical methods. Member of the EPFL AI Team's PLACE project (computer vision) and winner of the "Most Impactful Project" award at the PeaceTech hackathon. On academic leave this semester.

TECHNICAL SKILLS

- **Python** — object-oriented programming (encapsulation, inheritance, polymorphism), NumPy, Matplotlib, arcade, Jupyter
- **Web development** — TypeScript, React, Next.js; Python APIs with FastAPI, SQLAlchemy, PostgreSQL (SignalSafe project)
- **C++** — algorithms, recursion, data structures, complexity analysis
- **Applied linear algebra** — diagonalization, spectral theorem, least squares, singular value decomposition (SVD)
- **Analysis and numerical methods** — multivariable calculus, constrained optimization, Newton's method, numerical integration
- **Discrete mathematics** — combinatorics, graph theory
- **Tools** — Git / GitHub, LaTeX, Markdown, AI coding assistants; collaborative development and testing

PROJECTS

PLACE — EPFL AI Team | Computer vision · *since September 2026 — ongoing*

- Project team member: training models to geolocate a Street View image (coordinates, city or country).
- Scope: segmentation, object detection and classification, with an evaluation against human players.

PeaceTech Hackathon — ICRC challenge | **"Most Impactful Project" award** · *September 2026*

- **SignalSafe**: AI-assisted collection, deduplication and triage of content related to conflict-related sexual violence, for the ICRC.
- Contributed to the frontend (Next.js, React, TypeScript) and backend (FastAPI, SQLAlchemy, PostgreSQL) in a team of eight.

2D action-adventure video game | Python (arcade) · github.com/imarkria

- Object-oriented architecture designed from scratch (entities, map, engine and game loop), collision and interaction handling.
- Solo project taken through to a playable product, code versioned with Git.

Collaborative software project | Python · *pair project*

- Module allocation, testing and continuous integration of code with Git.

EXPERIENCE

Private mathematics tutor | Middle and high school students

Lausanne, Switzerland · Since 2024 — ongoing

- Diagnosing where students are stuck, tailoring explanations to their level, long-term follow-up.
- Fully autonomous in preparing sessions and managing the relationship with families.

Event steward and market vendor | Horse racing meetings and local market

Christchurch (NZ) · school exchange year · October 2019 – March 2020, every Sunday

- Worked independently across several roles over six months: ticketing, bar, sales and cash handling, horse care.

EDUCATION

EPFL — Bachelor in Mathematics | 2024 – present · *Lausanne, Switzerland*

CPGE PCSI — Lycée Henri-IV | 2023 – 2024 · *French preparatory classes in science and engineering · Paris, France*

French Baccalauréat, International Option (American OIB) — Highest honours (Mention Très Bien) | 2023

Lycée Jeannine Manuel, Lille, France · 20/20 in mathematics and physics

LANGUAGES

- **French** — native (C2) · **Spanish** — intermediate (B1)
- **English** — bilingual (C2): a year of immersion in New Zealand, American International Option baccalauréat

INTERESTS

- Horse riding · hiking · reading